

Second-order audit of the T6-entry composition: the small-amplitude interval question and the named verification item G-U1-SMALLT

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

Unit U1 assembled the candidate T6 theorem by composing the Prop-A floor with the STEP-5B budget: $\Delta F \geq 0.00432(1 - 1/\rho)$. This audit re-examines the composition's interval structure and finds ONE objection that must be UPHELD pending verification. Per the second-order-audit discipline, the finding is recorded with its named verification item, and the U1 proposal is flagged accordingly.

2. The finding

Prop A's proof is a four-interval decomposition in the trial mass (M -space). The $+0.00432$ figure is the BAND interval's worst-case sum (floor $P_B(M_+) = 0.0052460$ minus the sextic-only dip 9.2595×10^{-4}). But on interval I' ($M \geq M_+$, which CONTAINS the gap point $M_R = 0.109414$), the proof gives strict positivity with NO uniform margin: all Φ -coefficients are positive and $\Delta F_0 \geq 0$, with BOTH vanishing at the disordered point ($t = 0, m = m^*$). Near that point the layer excess of a shell competitor scales as

$$\Delta F_{\text{layer}} \approx \frac{\hat{\kappa}}{2} t^2 + \Delta F_0(m), \quad \hat{\kappa}|_{M=M_R} = \hat{r}_0(M_R) = m^* = 0.3045,$$

i.e. $O(t^2)$, while the certified beyond-layer envelope scales as $O(I^2)$ ($w_t \propto \lambda' I$) with the class-wide prefactor. If the chain's certified intensities I identify with the competitor amplitude scale t^2 , the comparison at fixed I is

$$\underbrace{\frac{m^*}{2} I - O(I^2)}_{\text{near-gap layer floor}} \quad \text{vs} \quad \underbrace{\frac{K(n_{\text{pack}})(\lambda' I)^2 J_{\text{eff}}}{4(1 - a_0)}}_{\text{certified envelope}},$$

with crossover $I_\times \approx (m^*/2)/c_{\text{env}}$, $c_{\text{env}} = \text{envelope}/I^2 \approx 4.3 \times 10^2$: numerically $I_\times \approx 3.5 \times 10^{-4}$ — INSIDE the certified window $[4 \times 10^{-4}, 2 \times 10^{-3}]$. At face value, then, the $0.00432(1 - 1/\rho)$ composition does NOT cover near-gap small-amplitude competitors at the certified intensities; the U1 arithmetic is valid only where the band floor is the binding layer protection.

3. Why this may be benign (two candidate resolutions)

- **Route (a) – self-consistent slaving + Hessian positivity:** physical competitors are evaluated at their self-consistent dressing (matched bookkeeping); near the gap point the off-diagonal operator enters the HESSIAN, where the P^2 theorem ($D + W = D_0 + \lambda' P^2 \geq D_0 >$

0) protects the quadratic order STRUCTURALLY — the perturbative $O(I^2)$ subtraction is then the wrong accounting in this regime (the correction renormalizes the positive quadratic form rather than competing with the floor). The verification this route requires: a one-page statement that the second-order envelope formula’s domain of use in the chain excludes (or structurally dominates) the near-gap regime, with the P^2 -positivity carrying that regime.

- **Route (b) – crossover arithmetic with per-pattern prefactor:** below the crossover the $O(I)$ floor dominates; at the certified intensities the relevant prefactor for SMALL patterns is the single-circle $K = 14$ (not the packing-maximal 107), shrinking c_{env} by $\times 7.6$ and pushing $I_{\times}$ to $\approx 2.7 \times 10^{-3} > 2 \times 10^{-3}$ — the window closes. The verification this route requires: the explicit inequality $\frac{m^*}{2}I - C_2^{-1}O(I^2) > K(n_{\text{eff}}(I))(\lambda'I)^2J_{\text{eff}}/(4(1-a_0))$ across the window, with the ΔF_0 curvature constant C_2 at the gap point (one new certified number) and the honest n_{eff} accounting (large- n patterns carry larger K but their near-gap layer floor also grows — the $\hat{\kappa}$ of a large star is bounded below by the same $\hat{r}_0(M)$, so the trade-off must be written carefully).

Either route, if verified, restores the U1 composition with an amended proof text (the band arithmetic for the band regime; the near-gap statement for the small-amplitude regime). Neither is verifiable from the migrated notes alone — the production bookkeeping’s calibration of “certified intensity” against the trial- (t, m) plane is the missing input.

4. Registered item and the U1 flag

- **G-U1-SMALLT (named verification item):** determine the layer floor binding at the certified competitor intensities — specifically (i) the calibration $I \leftrightarrow (t, m)$ of the production bookkeeping; (ii) the ΔF_0 gap-point curvature C_2 ; (iii) whichever of route (a)/(b) the chain’s structure supports, written as a one-page lemma. Until closed, the U1 composition’s quantifier “for every pattern of the amended class” is UNVERIFIED in the near-gap small-amplitude regime.
- **U1 proposal flag:** the T5 $\rightarrow$ T6-CONDITIONAL proposal of U1 is hereby amended to CONDITIONAL ON G-U1-SMALLT in addition to operator sign-off. The B1 status.json scope/notes are updated in this unit’s atomic set.
- What is NOT affected: B2 (Prop A itself is correct as stated — strict positivity holds; the issue is the COMPOSITION’s uniform-floor reading); B5 (the STEP-5B closure is a bound on the off-diagonal correction and stands within its own scope); the enumerated races (estimator-grade, per-pattern, unchanged).

5. Numerical verification

The audit’s arithmetic is estimate-grade on certified constants: $c_{\text{env}}(\text{endpoint}) = 1.74 \times 10^{-3}/(2 \times 10^{-3})^2 \approx 434$; $I_{\times} = (0.3045/2)/434 \approx 3.5 \times 10^{-4}$; route-(b) single-circle variant $c_{\text{env}}/7.6 \approx 57$, $I_{\times} \approx 2.7 \times 10^{-3}$. NO machine asserts executed this session; the crossover arithmetic is part of the G-U1-SMALLT item.

Quantitative sanity check (mandatory): consistency — at the LAM equilibrium ($I = 10^{-4}$, below $I_{\times}$ on either route) the artefact’s measured $\delta_{\text{LAM}} = 6.74 \times 10^{-8}$ sits five orders below the near-gap floor $(m^*/2) \times 10^{-4} = 1.5 \times 10^{-5}$: the ACTUAL corrections of raced patterns are deep inside the safe

region, consistent with the audit’s claim that the issue is the class-wide ENVELOPE’s coverage of a regime, not any observed counter-behavior; sign-direction — the audit errs conservative throughout (it treats the envelope as attainable, which the per-pattern measurements contradict by orders of magnitude); limit case — as $I \rightarrow 0$ the floor/envelope ratio diverges, so the flag concerns a bounded window, not the structure of the selection.

6. Devil’s-advocate (three objections; CLAUDE.md 6.3)

- (α) *“The audit invents a threat the operator-reviewed chain already considered: fourteen verdicts would have caught a regime gap.”* VALID-with-mitigation: very possibly — route (a) is likely the chain’s implicit structure (the P^2 theorem was BUILT to protect positivity structurally). But ‘likely implicit’ is not a citation: the U1 note COMPOSED two certified statements into a new theorem, and the composition’s validity in every regime is the U1 author’s burden, not the chain’s. The flag converts an implicit assumption into a named verifiable item — the cost is one page of verification; the alternative (an operator or referee finding it) is the documented failure mode this discipline exists to prevent.
- (β) *“The crossover arithmetic mixes normalizations: the audit itself admits uncertainty about whether certified I equals the trial t^2 .”* CORRECT — and that uncertainty IS the finding: a composition whose central inequality cannot be checked because the two sides’ amplitude calibrations are not pinned to each other in the migrated record is not sign-off-ready. G-U1-SMALLT item (i) is exactly this calibration.
- (γ) *“Flagging U1 as conditional-on-one-more-item after proposing it looks like instability.”* DISMISSED: it is the required behavior — the second-order audit (cross-turn discipline) exists to catch precisely this class of defect before operator review; the proposal’s substance (the assembly chain, the hypothesis enumeration, the band-regime arithmetic) stands; one regime’s coverage is added to the verification queue. An audit trail that shows the author catching their own gap is the intended state of the record.

7. Result footer

Result ID:	B1-RH-ENUM / t6-entry-composition-audit (unit U9; second-order audit of U1)
Precise statement:	AUDIT FINDING (UPHELD): the U1 composition’s uniform-floor reading of Prop A does not by itself cover near-gap small-amplitude competitors at the certified intensities (interval I’ floor is $O(I)$, envelope is $O(I^2)$ x large prefactor; estimated crossover $I_x \sim 3.5e-4$ inside the certified window). Two candidate resolutions recorded (P^2 structural route; crossover arithmetic with per-pattern prefactor + C_2 curvature). NAMED ITEM G-U1-SMALLT registered; the U1 T6-entry proposal is flagged CONDITIONAL on its closure.
Scope:	Audit of the U1 composition only. Prop A

(B2), the STEP-5B closure (B5), and the enumerated races (B1 T5) are unaffected within their own scopes.

Dependencies: U1 note; Math437 v1.2 interval proof; B5 artefact constants.

Evidence grade: AUDIT (estimate-grade crossover arithmetic; no machine asserts this session).

Reproduction command: (part of G-U1-SMALLT) crossover + C_2 arithmetic to be scripted with the U-series follow-up asserts.

Expected output: calibration statement; C_2 value; one of route (a)/(b) verified; amended U1 proof text.

Falsification gate: G-U1-SMALLT closing with NEITHER route verifiable (would demote the U1 proposal from conditional to withdrawn -- honest negative pre-registered).

Tier before / after: No tier action. B1 stays T5; the U1 proposal stands but now carries the G-U1-SMALLT condition.

No-overclaim statement: The audit does NOT assert the composition fails -- it asserts the composition is UNVERIFIED in one regime and names the verification. The crossover numbers are estimates. PDF build + linter PENDING operator-side.

Next required action: Close G-U1-SMALLT (production-bookkeeping calibration + C_2 + route lemma) BEFORE any operator sign-off on the U1 promotion. Mainline continues (U10) with this item at top priority.